# Solar Power Generation & Maintenance Advisor

## 1. Data Captured

The project collects solar energy generation, weather conditions, panel temperature, inverter fault details, and maintenance history to monitor solar panel performance.

## 2. Data Types and Columns

- Generation Data: Date, Time, Energy_kWh, Power_kW — records the energy produced and power output.
- Weather Data: Irradiance, Ambient_Temperature, Rainfall — captures environmental conditions affecting generation.
- Panel Data: Panel_Temperature, Voltage, Current — monitors the panel's operating conditions.
- Maintenance Data: Fault_Code, Maintenance_Date, Maintenance_Type — records faults and maintenance activities.

## 3. Analysis Done

- Descriptive Analytics: Analyzes daily and monthly energy generation, average output, and performance trends.
- Diagnostic Analytics: Investigates possible reasons for reduced generation by comparing output with weather and panel conditions.
- Predictive Analytics: Uses historical generation and weather data to forecast future solar energy production.
- Prescriptive Analytics: Recommends actions such as panel cleaning or inverter inspection to improve performance.

## 4. How Analysis Is Done

First, the data is cleaned to handle missing or incorrect values. Then, generation totals, averages, and shortfalls are calculated. Actual output is compared with expected output to investigate underperformance. Machine learning is used to forecast future generation, and the results help recommend suitable maintenance actions.

## 5. Tools Used

- Python: Performs data processing and analysis.

- Pandas: Cleans, organizes, and analyzes datasets.
- Matplotlib: Creates charts to visualize generation trends.
- Scikit-learn: Builds machine learning models to predict energy generation.
- Streamlit: Creates an interactive dashboard to display results.
- MySQL: Stores historical generation, weather, and maintenance data.